```
  set('testKeycloakVersion', "2.3.0")                    ❶
}

dependencies {
  ...
  testImplementation 'org.springframework.security:spring-security-test'
  testImplementation 'org.testcontainers:junit-jupiter'
  testImplementation "com.github.dasniko:
  ➥ testcontainers-keycloak:${testKeycloakVersion}"     ❷
}
```

❶ The version of Testcontainers Keycloak

❷ Provides Keycloak testing utilities on top of Testcontainers

We can implement full integration tests using `@SpringBootTest` and Testcontainers Keycloak. Since the setup is the same as for Catalog Service, I won't cover those tests here, but you can find them in the repository accompanying the book (Chapter12/12-end/order-service/src/test). Make sure you update those integration tests, or the application build will fail.

In this section, we'll test the web slice of a reactive application when the endpoints are secured, much like we did for Catalog Service.

TESTING SECURED REST CONTROLLERS WITH @WEBFLUXTEST AND SPRING SECURITY

We have already written autotests in `OrderControllerWebFluxTests` for the web slice using `@WebFluxTest`. Let's see now how to update them to take security into account.

As part of the OAuth2 Access Token validation, Spring Security relies on the public keys provided by Keycloak to verify the JWT signature. Internally, the framework configures a `ReactiveJwtDecoder` bean